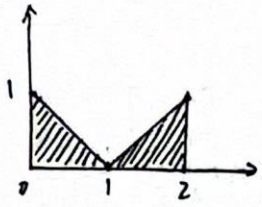
AT3007 Ass 2

Problem 1]

(a) True. For any $x, y \in (S_1 \cap S_2)$, we have $x, y \in S_1$, $x, y \in S_2$.
since S_1 is an nonempty convex set
then $\forall \lambda \in [0,1]$ $\lambda x + (1-\lambda) y \in S_1$
since S_2 is an nonempty convex set
then $\forall \lambda \in [0,1]$ $\lambda x + (1-\lambda) y \in S_2$
and since $x, y \in (S_1 \cap S_2)$
then $\forall \lambda \in [0,1]$ $\lambda x + (1-\lambda) y \in (S_1 \cap S_2)$

(b) False for $A_1 X \geq b_1$ $A_2 x \geq b_2$
if $A_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ -1 & -1 \end{bmatrix}$ $b_1 = \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix}$.

$A_2 = \begin{bmatrix} -1 & 0 \\ 0 & 1 \\ 1 & -1 \end{bmatrix}$ $b_2 = \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix}$



then the graph of the union is like which is not a polyhedron

(c) False $\min_{x \in \mathbb{R}^2} [1, 0]\, x$ s.t. $[1, 0]\, x \geq 1$, $x \geq 0$
then x is unbounded

(d) False if $m \leq \text{rank}(A) < n$, then the solution will not be consistant

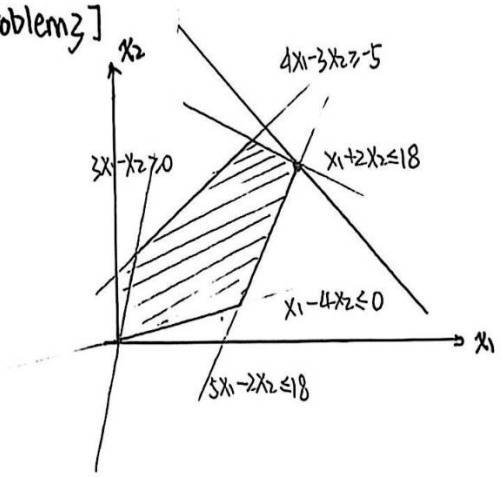
(e) True If there is more than one optimal solution, then understand in graph way, $c^T x$ will arrive at a line instead of a point to get the optimal solution, then there are infinitely many optimal solutions.

[Problem 2] minimize $y - x_2^+ + x_2^-$
$x_1^+, x_1^-, x_2^+, x_2^-, s_1, s_2, s_3, s_4, s_5, s_6, s_7, s_8, s_9, y, z$
subject to

$$y - x_1^+ + x_1^- - x_2^+ + x_2^- - x_3 - s_1 = 0 \quad (1)$$
$$y + x_1^+ - x_1^- + x_2^+ - x_2^- + x_3 - s_2 = 0 \quad (2)$$
$$x_1^+ - x_1^- - 1 - s_3 = -2 \quad (3)$$
$$x_1^+ - x_1^- - 1 + s_4 = 2 \quad (4)$$
$$z - x_2^+ + x_2^- - 1 - s_5 = 0 \quad (5)$$
$$z + x_2^+ - x_2^- + 1 - s_6 = 0 \quad (6)$$
$$z - 2x_3 + s_7 = 2 \quad (7)$$
$$x_3 + s_8 = 2 \quad (8)$$
$$x_1^+ - x_1^- - x_2^+ + x_2^- - x_3 - s_9 = 0 \quad (9)$$

$x_1^+, x_1^-, x_2^+, x_2^-, s_1, s_2, s_3, s_4, s_5, s_6, s_7, s_8, s_9, y, z \geq 0$

roblem3]



optimal solution : $x_1=x_2=6$ optimal value : 30

constraints are active: $x_1+2x_2\leq 18$

$5x_1-2x_2\leq 18$

vertices (0,0) (4,1) (6,6) (4,7) (1,3)

Problem4]

(a) given the vertices, we have the polyhedron

$$\begin{bmatrix} -4 & -1 \\ -4 & 3 \\ 0 & 1 \\ 2 & 5 \\ 1 & -2 \\ 3 & -8 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \leq \begin{bmatrix} -8 \\ 24 \\ 12 \\ 74 \\ 64 \\ 6 \end{bmatrix}$$

and set $A=\begin{bmatrix} -4 & -1 \\ -4 & 3 \\ 0 & 1 \\ 2 & 5 \\ 1 & -2 \\ 3 & -8 \end{bmatrix}$ $b=\begin{bmatrix} -8 \\ 24 \\ 12 \\ 74 \\ 64 \\ 6 \end{bmatrix}$

we suppose the upper left point of the square is (x_1, x_2), and the length of side is c

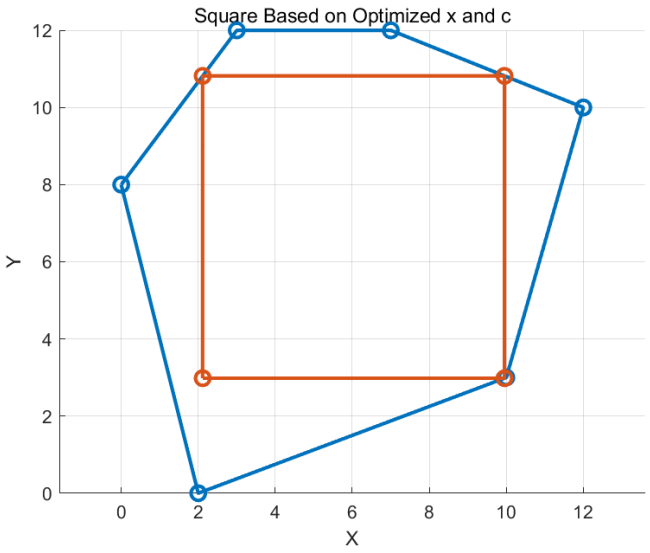
$$\begin{aligned} \text{maximize} \quad & c \\ x_1, x_2, c & \\ \text{subject to} \quad & A[x_1, x_2]^T \leq b \\ & A[x_1, x_2-c]^T \leq b \\ & A[x_1+c, x_2]^T \leq b \\ & A[x_1+c, x_2-c]^T \leq b \\ & x_1, x_2, c \geq 0 \end{aligned}$$

For objective function, to maximize the area is to maximize the length of side

For constraints, we must make sure that the square is entirely inside the polyhedron that is to make sure that the four vertices inside the polyhedron

(b) the optimal value is 7.83735 The largest area is 61.4303

The left upper point of the square is (2.1145, 10.8193)



[Problem 5]

(a) minimize $\sum_{i=1}^{m} y_i$
x_i, y_i, z_i

subject to
$y_i \geq a_i^T x - b_i \quad \forall i = 1, \dots, m$
$y_i \geq b_i - a_i^T x \quad \forall i = 1, \dots, m$
$z_i \geq x_i \quad \forall i = 1, \dots, m$
$z_i \geq -x_i \quad \forall i = 1, \dots, m$
$\sum_{i=1}^{n} z_i \leq \tau \quad \forall i = 1, \dots, m$

firstly I transform $|a_i^T x - b_i|$ into y_i, that is to make sure $y_i \geq \pm (a_i^T x - b_i)$

secondly transform $|x_i|$ into z_i that is to make sure $z_i \geq \pm x_i$

(b) $a_i = \begin{bmatrix} u_i \\ 1 \end{bmatrix} \quad b_i = v_i$

minimize $\sum_{i=1}^{m} |a_i^T x - b_i|$
$x \in \mathbb{R}^n$

subjective to $\|x\|_1 \leq \tau$

